Patients' Use and Evaluation of an Online System to Annotate Radiology Reports with Lay Language Definitions

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Rationale and Objectives: The increasing availability of personal health portals has made it easier for patients to obtain their imaging results online. However, the radiology report typically is designed to communicate findings and recommendations to the referring clinician, and may contain many terms unfamiliar to lay readers. We sought to evaluate a web-based interface that presented reports of knee MRI (magnetic resonance imaging) examinations with annotations that included patient-oriented definitions, anatomic illustrations, and hyperlinks to additional information.

Materials and Methods: During a 7-month observational trial, a statement added to all knee MRI reports invited patients to view their annotated report online. We tracked the number of patients who opened their reports, the terms they hovered over to view definitions, and the time hovering over each term. Patients who accessed their annotated reports were invited to complete a survey.

Results: Of 1138 knee MRI examinations during the trial period, 185 patients (16.3%) opened their report in the viewing portal. Of those, 141 (76%) hovered over at least one term to view its definition, and 121 patients (65%) viewed a mean of 27.5 terms per examination and spent an average of 3.5 minutes viewing those terms. Of the 22 patients who completed the survey, 77% agreed that the definitions helped them understand the report and 91% stated that the illustrations were helpful.

Conclusions: A system that provided definitions and illustrations of the medical and technical terms in radiology reports has potential to improve patients' understanding of their reports and their diagnoses.

Key Words: Radiology reports; lay language; patient-centered radiology; evaluation.

INTRODUCTION

The increasing availability of patient portals—personalized access to the electronic health record—has made it easier for patients to access appointment information, medication lists, and test results online (1–4). A survey of adult outpatients undergoing imaging within an academic health system revealed that 64% of respondents were interested in receiving an electronic copy of their radiology report in its original form (5). A cross-sectional study of more than 100,000 patients within a single health system demonstrated that 51% of the patients who had access to their radiology reports viewed their reports online (6).

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Radiology reports historically have been composed to communicate findings to the ordering physician and guide the next steps in a patient's management (7). Even though patients now can access their imaging reports, they often find it difficult to interpret the medical jargon contained in these results (8). In many cases, they also do not fully appreciate the rationale for undergoing an imaging test (9). Furthermore, the difficulty in understanding reports reviewed online in the absence of a discussion with a physician on the care team may result in miscommunication and unnecessary patient anxiety (10,11). To mitigate this anxiety, most patient portals release results after a fixed delay to allow ordering physicians to contact patients directly and discuss abnormal test results before patients can view the results online. A survey of referring physicians did not indicate that they experienced an increased workload as a result of the introduction of patient portals or the implementation of the delay in releasing results to patients (12).

Patients who send radiology-related messages to their providers through an electronic portal are most often interested in obtaining the results of their imaging examinations (13). As such, it is becoming increasingly important for radiologists to convey results of imaging examinations such that they are both appropriate for the ordering physician, but also consumable by the patient without a medical background. However, expecting two reports to now accompany every imaging study is not practical. We sought to understand patient sentiment about Patient-Oriented Radiology Reporter (PORTER), a web-based application to annotate reports, that was applied to reports generated for magnetic resonance imaging (MRI) of the knee (14). In this manuscript, we discuss the use and evaluation of this application by patients who accessed and reviewed their annotated reports through PORTER's web-based user interface.

METHODS

The organization's institutional review board approved the HIPAA (Health Insurance Portability and Accountability Act) compliant study protocol; informed consent was waived. From August 1, 2015 to February 29, 2016, we included a "Dear Patient" statement in the radiology department's default report template for knee MRI. The statement invited patients to view their report at the PORTER web address (Fig 1). The statement included the examination's accession number, which is an 8-digit number that uniquely identified each radiology examination in our department. Our institution does not consider the accession number to be protected health information. Patients used their examination's accession number, either through the online viewing portal or on printed copy of their report, in combination with their date of birth to gain access securely to the web site and view their knee MRI report.

When patients logged in during the trial period, the system recorded the patient's login time, retrieved their report, and searched for matching terms from a custom-built glossary of 310 terms (14). Each of the 190 primary terms included relevant abbreviations and alternate forms, and provided a lay language definition. When the user's mouse hovered over or clicked on an underlined term, PORTER displayed that term's definition as a pop-up balloon (Fig 2). We tallied the numbers of unique terms and total terms in each report; for example, if a report contained the term "medial meniscus" three times, that would count as one unique term and three total terms.

Alternate forms were shown with their primary term; for example, if the abbreviation "PCL" appeared in a report, the system displayed "posterior cruciate ligament (abbrev. PCL)." If available, a link to the corresponding Wikipedia page was displayed. If an image was available, it was displayed in a sidebar next to the report text and was updated as the user hovered over different terms. The system recorded the terms over which

Bones: Small tricompartmental osteophytes.
Soft Tissues: Mild prepatellar edema. There is also a multi loculated extra-articular fluid collection seen between the medial tibial plateau and the MCL, in keeping with bursitis.

DEAR PATIENT: Your knee MRI report is explained online at [redacted]. Enter this exam number ([redacted]) and your date of birth to view the explanation.

IMPRESSION:
1. Tricompartmental osteoarthritis, with moderate effusion and synovitis.

2. Tear of the posterior horn of the medial meniscus.

Figure 1. Part of a radiology report as shown in the patient portal. The web address and examination accession number have been obscured. MCL, medial collateral ligament; MRI, magnetic resonance imaging.

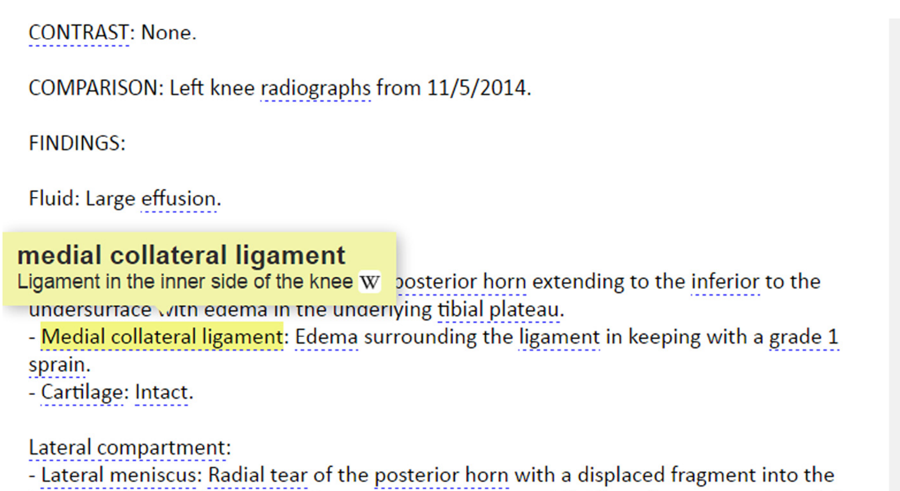

Figure 2. Part of a patient's radiology report as presented by PORTER (Patient-Oriented Radiology Reporter) shows the user hovering over a term ("medial collateral ligament"). The definition is shown in the yellow pop-up window; the "W" icon offers a link to a related Wikipedia page. A corresponding illustration appears on the right of the screen. (Color version of figure is available online.)

TABLE 1. Number of Terms Used in Reports and Viewed by the Patients

	Number Present in Report		Number Viewed		Fraction Viewed	
	Unique Terms	Total Terms	Unique Terms	Total Terms	Unique Terms (%)	Total Terms (%)
Minimum	31	44	0	0	0	0
Maximum	72	132	57	464	100	100
Mean	50.0	83.1	16.5	46.7	32	56
Median	50	82	15	23	28	28

Note: If the term "normal" appears four times in a report, for example, that is counted as one unique term and four total terms. The number viewed tallied all terms over which the user hovered to view the definition.

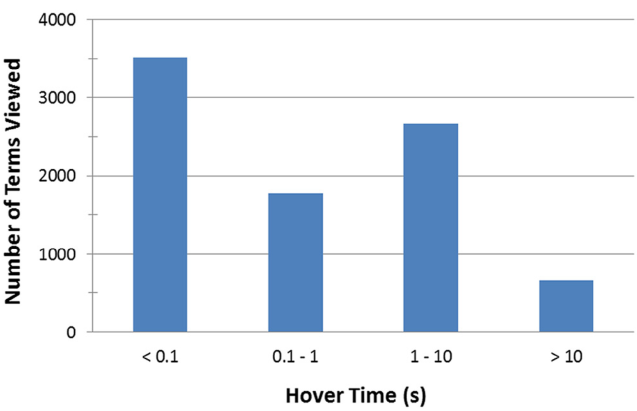

Figure 3. Histogram of the amount of time (in seconds) spent by patients hovering over each glossary term in their report.

TABLE 2. The 20 Most Commonly Viewed Terms and the Number of Times Viewed

Term	No. of Views
Normal	406
Cartilage	232
Medial meniscus	179
Medial collateral ligament	175
Signal	159
Posterior horn	142
Full thickness	133
Medial compartment	131
Patella	124
Lateral collateral ligament complex	124
Lateral meniscus	121
Subchondral	114
Edema	110
Anterior cruciate ligament	110
Knee effusion	91
Irregularity	86
Tricompartmental	83
Lateral femoral condyle	81
Popliteal cyst	80
Body	80

the patient hovered, the total time spent on the web site, and the amount of time spent on each term (Fig 3). To assess the total time spent viewing each report, we measured the elapsed time from each user's login to their last hover event. Gaps of more than 1 minute between events were excluded so that possible periods of inattention were not included.

The web site invited patients to complete a survey to share their opinions about the system and offered the chance to receive a $50 online gift card as an inducement. The survey consisted of six statements, such as "Reading my report online was helpful," which the patient could rate on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). We calculated the mean score for each statement and computed the level of agreement with each statement as the percentage of patients who responded either 4 (agree) or 5 (strongly agree). The survey allowed patients to enter free-text comments, and optionally to enter their email address if they wished to enter the drawing for the gift card.

RESULTS

During the 7-month trial, 185 of 1138 patients (16.3%) opened their report in the viewing portal. Of those, 136 patients (73.5%) opened the report once; seven patients opened their report four or more times. Of patients who logged in, 141 (76.2%) hovered over at least one term to view its definition and spent a mean of 3.2 minutes viewing their reports. The reports viewed contained a median of 50 unique terms and 82 total terms.

Patients viewed definitions of a median of 15 unique terms and 23 total terms per report; the mean, range, and median values are shown in Table 1. Patients hovered over a median of 28% of the unique terms and 28% of the total terms in a report. The median time spent hovering over each term was 200 ms. The mean time spent by each user hovering on terms in their reports was 2.4 minutes. In limiting the analysis to those terms hovered over for at least 1 second, 121 patients viewed a mean of 27.5 terms per examination and spent an average of 3.5 minutes viewing those terms. Table 2 lists the most commonly viewed terms.

Twenty-two of 185 patients (12%) completed the survey (Table 3); everyone who began the survey completed it, and all survey participants completed all questions. Respondents rated the information helpful to understand their report: the definitions of underlined words were helpful to 77% and pictures and drawings were helpful to 91%. Twelve individuals

TABLE 3. Survey Questions and Responses

Statement	Mean Score	Agree (%)
The definitions of underlined words helped me understand the report.	4.09	17 (77)
Reading my report online was helpful.	4.64	21 (95)
I understand better what radiologists do.	3.86	15 (68)
The pop-up definitions were distracting.	2.05	3 (14)
The pictures and drawings were helpful.	4.27	20 (91)
The definitions were easy to read and understand.	4.36	18 (82)

Note: Patients were asked to respond using a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The table shows the mean score and the number and percentage of patients who responded either 4 or 5 ("Agree").

provided free-text comments about the experience (Appendix); the remarks were generally favorable, and some suggested improvements such as using a larger font size for the image labels.

DISCUSSION

The Imaging 3.0 initiative sponsored by the American College of Radiology (ACR) recommends that the radiologists be present at every step of the imaging process: before, during, and after the examination (15). ACR also has convened a Commission on Patient- and Family-Centered Care to guide radiology practices to better adhere to these principles. PORTER was developed in an effort to empower patients to become more engaged in their care and to better consume the knowledge within their radiology reports. By providing definitions of medical and technical terms, relevant public-domain illustrations, and links to Wikipedia pages for additional information, we are attempting to demystify what has historically been a means of communication between physicians.

The study has several limitations. First, it focused exclusively on knee MRI examinations in outpatients at an academic medical center. It is likely that other patient subgroups may have different needs from a system that annotates their imaging results, and that this small pilot study has not captured those needs. In addition, the value of such a system to caregivers was not explored; this is a significant user base that may also have unique needs and interests. Work is under way to expand PORTER's glossary to cover all examinations performed in our department and to integrate the system with our health system's patient portal to simplify patients' access.

In light of evidence that 51% of patients with online access to radiology reports actually view their reports online (6), our findings that 18% of eligible patients used the system to view their reports are quite favorable. Our survey's 12% response rate is low, and further work is needed to engage more patients to evaluate the system, such as by offering simple icons for patients to register their favorable or unfavorable responses. A "splash screen" or other simple display could be used to inform patients about how to best use the annotated report.

The seemingly short median hover time may have reflected patients' attempts to explore the text and images associated with each of the underlined terms. The novelty of the system may have led them to hover over a variety of terms for short periods of time, possibly to scan the text of the definitions or to see which terms had associated images. The hovers of greater than 1 second in duration may present a more meaningful view of the users' experiences with the web site. Patients did not need to hover over report text to view their reports, so hover time and its related statistics likely underestimated the total time patients spent viewing their reports.

Patients value the opportunity to speak directly to a radiologist about their imaging results (9,16–18). Although PORTER was neither designed nor intended to replace conversations with a radiologist or follow up with the orthopedic specialists who ordered the knee MRIs, patients found value in the provided definitions and illustrations, and felt that they contributed to a better understanding of their diagnoses and imaging results. Of note, two patients mentioned that the annotated reports improved their understanding of the reports but not the next steps in their care (Appendix). Patients' comments suggest that they could be helped further by information that explains the causes and/or treatments of the reported findings. The comments also pointed up a need to assure that the system's images had easily readable anatomical labels and right/left labeling.

The current work represents an "InfoButton" approach: it provides links from an online text report to explanations of the report's terms and to publicly accessible resources on the Web (19,20). As medical care delivery evolves to become more patient centered and patient focused, the utility of patient portals will continue to increase. However, it is important to remain cognizant of the barriers that persist between patients and their health information, particularly when it comes to the consumption of radiology results (6,21,22). Radiology reports frequently include long sentences, complex polysyllabic technical terms, and unfamiliar vocabulary; they may be nearly impenetrable to the average patient despite the availability of online information resources that promote health literacy and engage patients in their own care (23). Patients' preferences to receive radiology results both in their original format and in a lay language summary should also be considered (5,24). Although investigators have explored systems to translate clinical text into lay language using an open-access, collaborative consumer health vocabulary, such systems are not yet part of mainstream practice (25). Systems such as PORTER have potential to help patients better understand their radiology results and the role of radiology and radiologists in the delivery of high-quality medical care.

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APPENDIX

Comments from patients who used the system.

- The popliteus myotendinous junction could use an explanatory image. Otherwise, great system!
- This was a really good idea, however, I have a strong medical background as a paramedic. For someone who does not "lay persons" they may panic over their results before they see their surgeon which could cause "hysteria" within the patient. I did love this though!
- Overall, I do not know what the test results actually mean, and overall, I do not understand what is wrong with my Knee in its entirety. I am waiting to speak to my specialist about the results and how I should proceed. . .. Reading the results on line gave me very little understanding of what is wrong!
- I didn't realize I could click on words for definitions but will do so in future. The illustration was extremely helpful. [BLINDED] gets an A for consumer friendliness.
- This system is fantastic! I understood the report perfectly. This is very helpful for individuals who were not educated in the field of medicine.
- The pictures and drawings were definitely helpful, but I couldn't read the script next to the pictures. It was too small and blurry.
- It would be helpful to explain the report in layman's terms.
- It would be helpful to have some indication of "left" and "right" on the drawings, or if I am looking "up" or "down" at the pictures. Also, some of the labels on the pictures were quite small and difficult for senior citizens to read.
- While reading the exam report (with plain English definitions) was helpful, it seems as if most of the report indicated that I had swelling of the knee, which I already knew, without any explanation or indication of why or what can be done. I am not sure that this plain English translation of the report prepared me to have a more in-depth and knowledgeable conversation with my orthopedist.
- I think this is a great service for patients who have no anatomical knowledge. I found it very interesting and helpful in understanding my diagnosis.
- The information was helpful. If there is a way for you to provide the information with pictures and drawings of my condition, that would be awesome!
- I almost missed the underlines. I am using a nexus 10 tablet. I read report, started survey and switched back to report to use the helpful diagrams and underlines. Great job, thanks for helping understand. I was reading a knee MRI, which is very complex.